



INDIAN INSTITUTE OF SPACE SCIENCE AND
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PROJECT REPORT

Hyperspectral Image Super-Resolution using VMamba-based Pansharpening

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December 17 2025

Declaration

I hereby declare that the project report titled "***Hyperspectral Image Super-Resolution Using VMamba-based Pansharpening***" submitted in partial fulfillment of the requirements for the award of the degree of **Master of Technology (M.Tech) in Machine Learning and Computing** is a record of the original work carried out by me under the supervision of **Dr. Deepak Mishra**. This work has not been submitted previously, in part or full, for the award of any degree, diploma, associateship, fellowship, or other similar title in any other Institution or University. Proper acknowledgments have been made wherever the work of others has been cited, in keeping with academic and ethical standards.

Place: Thiruvananthapuram

Date: December 17, 2025

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Certificate

This is to certify that the project report titled "***Hyperspectral Image Super-Resolution using VMamba-based Pansharpening***" submitted by **Sai Kumar Reddy**, to the **Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram**, in partial fulfillment of the requirements for the award of the degree of **Master of Technology (M.Tech) in Machine Learning and Computing**, is a bona fide record of the original work carried out by him/her under my supervision. The contents of this project report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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Abstract

Hyperspectral imaging provides rich spectral information crucial for remote sensing applications but suffers from limited spatial resolution due to fundamental sensor constraints. Pansharpening addresses this limitation by fusing low-resolution hyperspectral images (LR-HSI) with high-resolution panchromatic images (HR-PAN) to generate high-resolution hyperspectral outputs (HR-HSI). This work presents a novel deep learning approach leveraging Vision Mamba (VMamba), a state space model architecture, for efficient and high-quality hyperspectral pansharpening.

Traditional pansharpening methods including component substitution and multi-resolution analysis techniques introduce spectral distortions and rely on fixed fusion rules. While deep learning methods have shown improvements, CNNs are limited by local receptive fields, and Transformers, despite achieving excellent results, suffer from quadratic $O(N^2)$ computational complexity, making them impractical for large-scale satellite image processing.

The proposed VMamba-based architecture employs a dual-encoder design to independently process hyperspectral and panchromatic inputs, followed by cross-attention fusion for adaptive feature integration. A hierarchical four-stage VMamba backbone with 2D Selective-Scan (SS2D) provides global receptive fields while maintaining linear $O(N)$ complexity. The system is trained using a composite loss function combining ℓ_1 reconstruction, Spectral Angle Mapper (SAM), edge preservation, and Structural Similarity Index Measure (SSIM) losses to balance spatial sharpness and spectral fidelity.

Keywords: Hyperspectral imaging, Pansharpening, Image fusion, State space models, VMamba, Deep learning, Remote sensing, Computer vision.

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Chapter 1

Introduction

1.1 Motivation and Background

The ability to identify and classify materials based on their spectral signatures has revolutionized Earth observation and remote sensing. Hyperspectral imaging technology, which captures electromagnetic radiation across hundreds of narrow, contiguous spectral bands, has emerged as a powerful tool enabling precise discrimination between materials that appear identical to the human eye or conventional RGB cameras. This capability stems from the fundamental principle that different materials absorb and reflect electromagnetic radiation differently across the spectrum, creating unique spectral signatures—essentially “fingerprints” that can be used for identification and classification.

Applications of hyperspectral imaging span diverse domains with significant societal impact. In **precision agriculture**, hyperspectral sensors detect subtle changes in plant physiology indicating disease, nutrient deficiency, or water stress before visible symptoms appear, enabling targeted interventions that reduce chemical usage and increase crop yields. For **environmental monitoring**, these sensors track water quality in lakes and coastal regions by identifying algal blooms and pollutants, monitor forest health and species composition, and assess ecosystem changes in response to climate variations. In **mineral exploration**, hyperspectral imaging identifies ore deposits and geological formations through characteristic spectral absorption features, reducing exploration costs and environmental impact. Urban planning, disaster response, and national security applications further demonstrate the technology’s versatility and importance.

However, hyperspectral imaging faces a fundamental physical constraint that limits its effectiveness: the inverse relationship between spatial and spectral resolution. To capture detailed spectral information across numerous narrow bands, sensors must integrate light over longer exposure times or larger ground sampling areas, inherently reducing spatial resolution. This trade-off arises from basic principles of optics and detector physics—dividing incident light into many spectral channels reduces the photon flux available to each detector element, necessitating larger collection areas or longer integration times to maintain acceptable signal-to-noise ratios.

1.1.1 The Spatial-Spectral Resolution Trade-off

Consider a practical example: The AVIRIS (Airborne Visible/Infrared Imaging Spectrometer) sensor captures 224 spectral bands covering the range from 400 to 2500 nm with 10 nm spectral resolution, providing exceptional spectral detail. However, from typ-

ical aircraft altitudes of 4-20 km, AVIRIS achieves spatial resolutions of only 4-20 meters per pixel. Similarly, the spaceborne Hyperion sensor on NASA’s EO-1 satellite provides 220 spectral bands but at 30-meter spatial resolution. While adequate for large-scale phenomena like forest classification or geological mapping, this resolution proves insufficient for many applications requiring both detailed spectral information and fine spatial detail.

Agricultural applications illustrate this limitation clearly. Identifying specific crop diseases requires detailed spectral signatures to discriminate between healthy tissue, various disease states, and nutritional deficiencies. However, early detection demands identification of small infected areas, potentially just meters across in large fields. Current hyperspectral sensors force a choice: either sacrifice spectral detail for spatial resolution (using multispectral imagers with 5-10 broad bands at high spatial resolution) or accept coarse spatial sampling to maintain spectral richness. Neither option is optimal for precision agriculture, environmental monitoring at fine scales, or urban analysis requiring material identification of small objects.

1.2 Pansharpening: Bridging the Resolution Gap

Pansharpening offers an elegant solution to this fundamental constraint by leveraging the complementary strengths of two sensor types that can be co-located on the same platform or acquired from different sources:

1. **Low-Resolution Hyperspectral Image (LR-HSI):** Provides rich spectral information with 100+ bands but limited spatial detail (e.g., 30-meter resolution)
2. **High-Resolution Panchromatic Image (HR-PAN):** Captures a single broad spectral band (typically 400-900 nm) but with fine spatial detail (e.g., sub-meter to a few meters resolution)

The pansharpening process fuses these complementary data sources to generate a high-resolution hyperspectral image (HR-HSI) that ideally combines the spectral richness of the LR-HSI with the spatial sharpness of the HR-PAN. This fusion can be conceptualized as answering the question: "Given that we know the approximate spectral composition at coarse resolution and the precise spatial structure at high resolution, can we infer what the detailed spectral composition would be at high spatial resolution?"

1.2.1 Mathematical Formulation

Formally, the pansharpening problem can be stated as learning a mapping function $\mathcal{F}$:

$$\mathcal{F} : (\mathbf{H}_{LR}, \mathbf{P}_{HR}) \rightarrow \mathbf{H}_{HR} \quad (1.1)$$

where:

- $\mathbf{H}_{LR} \in \mathbb{R}^{h \times w \times c}$ is the low-resolution hyperspectral image with spatial dimensions $h \times w$ and c spectral bands
- $\mathbf{P}_{HR} \in \mathbb{R}^{rh \times rw}$ is the high-resolution panchromatic image with spatial scaling factor r (typically $r = 4$)

- $\mathbf{H}_{HR} \in \mathbb{R}^{rh \times rw \times c}$ is the desired high-resolution hyperspectral image

The quality of the pansharpened image $\mathbf{H}_{HR}$ must satisfy two critical requirements:

Spatial Consistency: When spatially degraded (downsampled) to the original low resolution, $\mathbf{H}_{HR}$ should match $\mathbf{H}_{LR}$:

$$\text{Downsample}(\mathbf{H}_{HR}) \approx \mathbf{H}_{LR} \quad (1.2)$$

Spectral Consistency: When spectrally averaged to create a panchromatic-like image, $\mathbf{H}_{HR}$ should match $\mathbf{P}_{HR}$:

$$\text{SpectralAverage}(\mathbf{H}_{HR}) \approx \mathbf{P}_{HR} \quad (1.3)$$

These constraints ensure that the pansharpened result is consistent with both input sources while enhancing resolution.

1.3 Challenges in Hyperspectral Pansharpening

Despite decades of research, hyperspectral pansharpening remains challenging due to several fundamental difficulties:

1.3.1 Spectral-Spatial Trade-off

The most critical challenge is balancing spectral fidelity with spatial enhancement. Aggressive spatial detail injection from the panchromatic image can distort spectral signatures, potentially causing misclassification of materials. Conversely, prioritizing spectral preservation may result in insufficient spatial enhancement, defeating the purpose of pansharpening. This trade-off is particularly acute for hyperspectral data where spectral accuracy is often more critical than for conventional multispectral imaging.

1.3.2 High Dimensionality

Hyperspectral images typically contain 100-200+ spectral bands, creating substantial computational and statistical challenges. Traditional methods designed for 3-4 band RGB or multispectral images often fail to scale effectively to hyperspectral dimensionality. The high-dimensional feature space exacerbates issues like the curse of dimensionality, requiring more sophisticated approaches to capture inter-band correlations and spectral-spatial relationships.

1.3.3 Limited Training Data

Deep learning methods require paired training data: (LR-HSI, HR-PAN, HR-HSI) triplets. However, true HR-HSI is rarely available since acquiring it would require the very technology pansharpening aims to simulate. Researchers typically rely on Wald's protocol, which synthesizes training data by degrading existing images. While practical, this approach creates a domain gap between synthetic training data and real operational scenarios where the degradation process may differ from the simulation.

1.3.4 Sensor Variations

Different sensor platforms have varying spectral response functions, spatial resolutions, and noise characteristics. Models trained on data from one sensor may generalize poorly to others, requiring either sensor-specific training or domain adaptation techniques. This limits the practical deployment of pansharpening algorithms across diverse satellite and airborne platforms.

1.4 Evolution of Pansharpening Approaches

1.4.1 Traditional Methods: Signal Processing Era

Early pansharpening research focused on signal processing techniques that, while computationally efficient, employed fixed fusion rules unable to adapt to scene content.

Component Substitution (CS) methods such as Intensity-Hue-Saturation (IHS) , Principal Component Analysis (PCA) , and Gram-Schmidt (GS) orthogonalization transform the multispectral image into an alternative representation, replace one component with the panchromatic image, and transform back. These methods are simple and fast but introduce severe spectral distortions, particularly for hyperspectral data with many bands where the assumptions underlying these transforms break down.

Multi-Resolution Analysis (MRA) methods including wavelet transforms , contourlet transforms , and Laplacian pyramids decompose images into multiple frequency bands and inject high-frequency details from the panchromatic image into the upsampled hyperspectral image. MRA methods generally preserve spectral information better than CS approaches but can produce spatial artifacts and have limited ability to model complex scene-dependent fusion strategies. The MTF-GLP method , which incorporates sensor modulation transfer functions, represents the state-of-the-art among traditional approaches but still falls short of deep learning methods in overall quality.

1.4.2 Deep Learning Revolution

The application of deep learning to pansharpening marked a paradigm shift, enabling end-to-end learning of fusion strategies from data rather than relying on hand-crafted rules.

CNN-based pioneers like PNN demonstrated that even simple three-layer convolutional networks could outperform traditional methods. PanNet introduced residual learning specifically for pansharpening, learning to predict the high-frequency detail layer rather than the complete image, accelerating convergence and improving results. DRPNN employed deeply recursive networks with parameter sharing, enabling very deep architectures without proportional parameter growth.

However, CNN-based methods face a fundamental limitation: **restricted receptive fields**. Even deep networks struggle to capture global context, limiting their ability to model long-range spatial-spectral dependencies. For hyperspectral data where spectral correlations can span distant wavelengths and spatial patterns may extend across large areas, this local processing paradigm proves suboptimal.

Transformer-based methods addressed the receptive field limitation through self-attention mechanisms. SpectralFormer and PanFormer apply vision transformers to pansharpening, achieving excellent results through global context modeling. However,

transformers suffer from **quadratic computational complexity** $O(N^2)$ where N is the number of image patches, making them impractical for large satellite scenes. For a typical hyperspectral image of 1000×1000 pixels divided into patches, the attention matrix alone requires gigabytes of memory, and processing time scales poorly.

1.5 State Space Models: A New Paradigm

Recent advances in sequence modeling have introduced State Space Models (SSMs) as a compelling alternative that combines the global receptive field of transformers with the linear complexity of CNNs.

1.5.1 From Mamba to VMamba

Mamba, building on earlier work with S4 models introduced selective state space mechanisms where transition parameters adapt based on input content:

$$\mathbf{h}_t = \mathbf{A}(x_t)\mathbf{h}_{t-1} + \mathbf{B}(x_t)\mathbf{x}_t \quad (1.4)$$

$$\mathbf{y}_t = \mathbf{C}(x_t)\mathbf{h}_t \quad (1.5)$$

This selectivity enables the model to focus on relevant information while maintaining linear $O(N)$ complexity through efficient implementations using hardware-aware algorithms.

VMamba [e]xtended Mamba to 2D vision tasks through **2D-Selective-Scan (SS2D)**, which processes images by scanning in four directions (left-to-right, right-to-left, top-to-bottom, bottom-to-top), capturing spatial relationships in all orientations without directional bias. Each directional scan flattens the 2D image into a 1D sequence, processes it through the selective SSM, and reshapes the output back to 2D. The four directional outputs are then merged, providing comprehensive spatial coverage with global receptive fields while maintaining linear complexity.

1.6 Research Objectives and Contributions

This project investigates the application of VMamba to hyperspectral pansharpening, hypothesizing that its unique combination of global context modeling and computational efficiency makes it ideal for this task.

1.6.1 Primary Objectives

1. **Architecture Development:** Design a VMamba-based pansharpening architecture that effectively processes hyperspectral and panchromatic inputs, incorporating dual encoders for independent feature extraction, cross-attention fusion for adaptive integration, and a hierarchical VMamba backbone for multi-scale processing.
2. **Training Strategy:** Develop a robust training methodology including composite loss functions balancing spatial and spectral quality, data augmentation techniques suitable for hyperspectral data, and regularization strategies ensuring generalization.

3. **Comprehensive Evaluation:** Conduct extensive experiments comparing against state-of-the-art methods across multiple quality metrics (PSNR, SSIM, SAM, ERGAS) and computational efficiency measures (parameters, FLOPs, inference speed).
4. **Ablation Analysis:** Systematically validate design choices through ablation studies isolating the contribution of each architectural component and loss function.

1.6.2 Key Contributions

This work makes several novel contributions to hyperspectral image processing:

1. **First VMamba Application to Pansharpening:** To the best of our knowledge, this represents the first application of Vision Mamba architecture to hyperspectral pansharpening, demonstrating that selective state space models can effectively address image fusion tasks.
2. **Efficient Architecture Design:** A carefully designed system achieving 71% parameter reduction (29.8M vs 102.7M parameters) and $3.5\times$ faster inference compared to transformer-based methods while delivering superior quality across all metrics.
3. **Cross-Attention Fusion:** A novel fusion module using cross-attention where panchromatic features query hyperspectral features to selectively preserve spectral information while injecting spatial details, enabling content-adaptive fusion superior to simple concatenation or addition.
4. **Comprehensive Validation:** Extensive experimental evaluation including ablation studies, computational analysis, and comparison with 9 baseline methods demonstrating the effectiveness and efficiency of the proposed approach.

1.7 Potential Impact and Applications

The developed VMamba-based pansharpening system has immediate practical applications:

Environmental Monitoring: Enhanced hyperspectral imagery enables more accurate tracking of deforestation, water quality degradation, and ecosystem changes with the spatial resolution needed to identify small-scale phenomena.

Precision Agriculture: Farmers can benefit from high-resolution hyperspectral data identifying crop diseases, nutritional deficiencies, and water stress in small field sections, enabling targeted interventions reducing chemical usage.

Mineral Exploration: Mining companies can use pansharpened hyperspectral imagery to identify ore deposits with higher spatial precision, reducing exploration costs and environmental impact.

Operational Deployment: The computational efficiency (linear complexity, reduced parameters) makes real-time or near-real-time processing feasible, enabling deployment on edge devices like drones and small satellites where computational resources are limited.

Chapter 2

Literature Review

2.1 Overview

Hyperspectral pansharpening has evolved significantly over the past three decades, progressing from classical signal processing techniques to sophisticated deep learning architectures. This chapter provides a comprehensive review of existing approaches, organized into four main categories: traditional signal processing methods, machine learning techniques, deep learning approaches, and state space models. For each category, we analyze the underlying principles, advantages, limitations, and suitability for hyperspectral data. This review identifies critical research gaps that motivate the proposed VMamba-based approach.

2.2 Traditional Signal Processing Methods

Traditional pansharpening methods, developed primarily in the 1990s and early 2000s, rely on hand-crafted fusion rules based on signal processing principles. While computationally efficient, these methods struggle with the high dimensionality and complexity of hyperspectral data.

2.2.1 Component Substitution Methods

Component Substitution (CS) methods transform multispectral images into alternative representations, replace one component with the panchromatic image, and perform inverse transformation.

Intensity-Hue-Saturation (IHS)

The IHS transform converts RGB images to intensity, hue, and saturation components. The intensity component is replaced with the histogram-matched panchromatic image, then the inverse transform generates the pansharpened result. While simple and fast ($O(N)$ complexity with no training required), IHS introduces severe spectral distortions when applied beyond three bands. For hyperspectral data with 100+ bands, IHS is fundamentally unsuitable as the color space transformation assumptions break down completely. Studies show PSNR degradation of 5-8 dB compared to modern methods and SAM errors exceeding 0.15 radians, indicating substantial spectral corruption.

Principal Component Analysis (PCA)

PCA-based pansharpening projects multispectral bands onto principal components ordered by variance, replaces the first component (containing maximum variance) with the panchromatic image, and reconstructs through inverse PCA. This method adapts better to scene content than IHS, as the principal components are data-driven rather than fixed. However, for hyperspectral data, several issues emerge: (1) the first principal component may not capture all spatial information, (2) computational cost scales with the square of the number of bands for covariance matrix computation, and (3) spectral distortions remain significant, particularly for materials with spectra poorly aligned with principal directions. Typical performance shows 2-3 dB PSNR improvement over IHS but still lags deep learning methods by 5-7 dB.

2.2.2 Multi-Resolution Analysis Methods

Multi-Resolution Analysis (MRA) methods decompose images into multiple scales and inject spatial details at appropriate frequency levels, generally providing better spectral preservation than component substitution approaches.

Wavelet-Based Fusion

Wavelet transforms decompose the panchromatic image into approximation (low-frequency) and detail (high-frequency) coefficients at multiple scales. High-frequency details are injected into each upsampled hyperspectral band, with coefficients often weighted based on local statistics. The à trous wavelet algorithm is particularly popular for pansharpening due to its shift-invariance property. Common wavelets include Haar (simplest but generates blocky artifacts), Daubechies (db4, db8 for smoother results), and Symlets (near-symmetric, reduced phase distortion).

Advantages include multi-scale processing capability and better spectral preservation than CS methods (typical SAM: 0.06-0.08 radians). However, limitations include: (1) limited directionality—standard 2D wavelets provide only three orientations (horizontal, vertical, diagonal), potentially missing edge orientations, (2) ringing artifacts near sharp edges due to the oscillatory nature of wavelet basis functions, and (3) fixed wavelet selection unable to adapt to varying image content. Performance is moderate, with PSNR values typically 35-37 dB on standard datasets, falling short of deep learning methods by 4-6 dB.

Contourlet and Curvelet Transforms

To address wavelets' limited directionality, contourlet and curvelet transforms provide multi-directional decomposition. Contourlets combine Laplacian pyramids for multi-scale analysis with directional filter banks, capturing anisotropic features like edges and contours more effectively. Curvelets extend this with optimal representation of curve-like singularities.

These methods show improvement over wavelets for images with strong directional features, achieving PSNR gains of 0.5-1.0 dB. However, they introduce significant computational complexity (5-10× slower than wavelets), generate redundant representations increasing memory requirements, and remain inferior to deep learning approaches in overall quality metrics. The fixed multi-directional decomposition, while better than wavelets,

still cannot match the adaptive feature extraction of learned methods.

MTF-Based Methods

Methods incorporating Modulation Transfer Function (MTF) modeling [aiazzi2006mtf] account for sensor optical characteristics in the fusion process. MTF-GLP (Generalized Laplacian Pyramid) represents the state-of-the-art among traditional methods, using MTF-matched filters in pyramid decomposition for physically accurate fusion.

MTF-GLP achieves the best performance among traditional approaches: PSNR 37-38 dB, SSIM 0.94-0.95, SAM 0.05-0.07 radians on standard datasets. The physically motivated design provides better balance between spatial enhancement and spectral preservation. However, limitations include: (1) requirement for accurate MTF knowledge, often unavailable or imperfectly known for operational sensors, (2) computational complexity moderate but higher than simple CS methods, (3) fixed fusion strategy unable to adapt to scene-specific characteristics, and (4) performance gap of 3-4 dB PSNR compared to modern deep learning methods.

2.2.3 Summary of Traditional Methods

Traditional methods offer computational efficiency and no training data requirements, making them suitable for quick processing or scenarios with limited computational resources. However, fundamental limitations constrain their performance: (1) fixed, hand-crafted rules unable to learn optimal fusion strategies from data, (2) inability to model complex, non-linear relationships between spatial and spectral information, (3) poor scalability to high-dimensional hyperspectral data, and (4) difficulty achieving optimal spectral-spatial quality trade-off. These limitations motivated the shift toward data-driven learning approaches.

2.3 Deep Learning Methods

The application of deep learning to pansharpening marked a paradigm shift, enabling end-to-end learning of fusion strategies from data. Deep learning methods consistently outperform traditional approaches but introduce new challenges related to architecture design, training requirements, and computational efficiency.

2.3.1 CNN-Based Approaches

Convolutional Neural Networks represent the first wave of deep learning applied to pansharpening, demonstrating that learned features outperform hand-crafted fusion rules.

PNN: Pioneer Deep Learning Method

Pansharpening Neural Network (PNN) introduced deep learning to this domain with a simple three-layer architecture: input layer accepting concatenated upsampled hyperspectral and panchromatic images, two convolutional layers (64 filters 9×9 , then 32 filters 5×5) with ReLU activation, and output layer generating pansharpened result. Despite architectural simplicity (only 500K parameters), PNN outperformed traditional methods by 2-3 dB PSNR, demonstrating the power of learned feature extraction.

However, PNN’s shallow architecture limits representational capacity. The large receptive fields (9×9 , 5×5) were necessary to capture sufficient context given limited depth, but this creates computational inefficiency and difficulty training deep variants. No residual connections makes optimization challenging, and the simple concatenation fusion strategy treats all input information equally rather than selectively combining features. Nevertheless, PNN established the feasibility and advantages of end-to-end learned pansharpening.

PanNet: Residual Learning Breakthrough

PanNet [yang2017pannet] introduced two key innovations: residual learning and high-pass filtering. Rather than predicting the complete pansharpened image, PanNet learns the high-frequency detail layer (difference between desired output and upsampled input), significantly simplifying the learning task. The panchromatic image undergoes high-pass filtering before network input, explicitly extracting spatial details for injection.

The architecture employs an 8-layer residual network with batch normalization for stable training. Each residual block contains Conv-BN-ReLU-Conv-BN with skip connections enabling gradient flow through deep networks. This design achieves 38-39 dB PSNR (3-4 dB improvement over PNN) with 5M parameters.

Advantages include faster convergence through residual learning, better spectral preservation by predicting only the detail layer, and stable training via batch normalization. Limitations include: (1) fixed high-pass filter not learned from data, potentially suboptimal for varying image content, (2) simple concatenation fusion lacking adaptive weighting, (3) local receptive field limited even with 8 layers, and (4) single-scale processing unable to handle objects of varying sizes optimally.

DRPNN: Deep Recursive Networks

Deeply Recursive Pansharpening Neural Network (DRPNN) addresses the depth-parameter trade-off through recursive units. A single convolutional block is applied recursively R times (typically $R=10-15$), with each iteration’s output feeding the next iteration’s input. Parameter sharing across iterations enables very deep processing (equivalent to 20-30 layer networks) with only 8M parameters.

Each recursive unit consists of Conv-ReLU-Conv with skip connections. Global skip connections from input to output preserve low-frequency information. This design achieves 39-40 dB PSNR with strong feature extraction capability. However, limitations include: (1) sequential recursive processing creates computational bottlenecks less amenable to parallelization, (2) gradient flow issues if recursion depth R is large, requiring careful initialization, (3) all recursive iterations share identical parameters, limiting representational flexibility, and (4) receptive field remains local despite depth.

2.3.2 Attention Mechanisms and Multi-Scale Processing

Later CNN-based methods incorporated attention mechanisms and multi-scale architectures to address limitations of early approaches.

Multi-Scale Fusion Networks

FusionNet and related architectures process features at multiple resolutions: Path 1 operates at original resolution, Path 2 downsamples by $2\times$, Path 3 downsamples by $4\times$. Each path extracts features via convolutional blocks, capturing patterns at different scales. Features are concatenated and passed through attention modules weighting important patterns. The decoder progressively upsamples with skip connections from encoder paths preserving fine details.

Multi-scale processing provides better handling of objects of varying sizes, with attention focusing on relevant features. Performance reaches 39-40 dB PSNR. However, computational cost is heavy (23M parameters, 156 GFLOPs), requiring substantial GPU memory (7+ GB). The architecture complexity increases training difficulty and data requirements. Despite improvements, the fundamental CNN limitation of local receptive fields persists—even with multi-scale processing, capturing long-range dependencies remains challenging.

2.3.3 GAN-Based Approaches

Generative Adversarial Networks introduce adversarial training for enhanced perceptual quality, using a generator to create pansharpened images and a discriminator to distinguish real from generated outputs.

PSGAN Architecture

PSGAN (Pansharpening GAN) employs a U-Net-style generator with encoder-decoder architecture and skip connections. The encoder extracts multi-scale features from concatenated HSI and PAN inputs through progressive downsampling. The decoder reconstructs high-resolution output via transposed convolutions with attention mechanisms weighting spatial-spectral features. The discriminator uses multi-scale discrimination analyzing images at multiple resolutions to detect artifacts at different scales.

The loss function combines adversarial loss encouraging realistic outputs, L1 reconstruction loss for pixel accuracy, and spectral loss (SAM) for spectral fidelity:

$$\mathcal{L}_{total} = \lambda_{adv}\mathcal{L}_{adversarial} + \lambda_{L1}\mathcal{L}_{L1} + \lambda_{SAM}\mathcal{L}_{SAM} \quad (2.1)$$

PSGAN achieves excellent perceptual quality with sharp spatial details (PSNR 40-41 dB) and visually pleasing results. However, significant challenges include: (1) training instability—GANs are notoriously difficult to train, prone to mode collapse where the generator produces limited diversity, and oscillation where training fails to converge, (2) spectral artifacts—adversarial training prioritizing perceptual quality can introduce subtle spectral distortions, (3) hyperparameter sensitivity—loss weight balancing requires extensive tuning, and (4) computational cost—training requires $2\times$ GPU memory (generator + discriminator) and 2-3 $\times$ training time compared to non-adversarial methods.

Despite strong results, GAN instability and spectral accuracy concerns limit adoption for hyperspectral applications where spectral fidelity is critical. The method works well for RGB/multispectral pansharpening but struggles with high-dimensional hyperspectral data where the discriminator’s task becomes increasingly complex.

2.3.4 Transformer-Based Methods

Transformers revolutionized natural language processing and computer vision through self-attention mechanisms capturing long-range dependencies. Their application to pansharpening addresses CNN receptive field limitations but introduces computational challenges.

Self-Attention for Global Context

Vision Transformers (ViT) divide images into patches, treat them as tokens, and apply multi-head self-attention:

$$\mathbf{Q} = \mathbf{XW}_Q, \quad \mathbf{K} = \mathbf{XW}_K, \quad \mathbf{V} = \mathbf{XW}_V \quad (2.2)$$

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_k}}\right) \mathbf{V} \quad (2.3)$$

This mechanism computes relationships between all patch pairs, providing global receptive fields from the first layer—a fundamental advantage over CNNs requiring depth for large receptive fields.

SpectralFormer

SpectralFormer adapts Vision Transformers for hyperspectral classification and has been extended to pansharpening. The architecture employs patch embedding dividing images into non-overlapping patches, positional encoding adding spatial information, and a transformer encoder with L layers (typically $L=12$) each containing multi-head self-attention ($h=8$ heads) and feed-forward networks (MLP with GELU activation).

For pansharpening, separate encoders process HSI and PAN patches, fusion occurs via attention mechanisms, and the decoder reconstructs high-resolution output. Performance is excellent: PSNR 40-41 dB, SSIM 0.968-0.970, SAM 0.040-0.045 radians, representing state-of-the-art quality circa 2022-2023.

However, critical limitations constrain practical deployment: (1) **Quadratic complexity**—For N patches, self-attention requires $O(N^2)$ operations and memory. A 256×256 image with 16×16 patches yields $N=256$, requiring $256^2=65,536$ attention computations per layer. With 12 layers, this becomes computationally prohibitive for large scenes. (2) **Massive parameters**—Typical implementations use 100M+ parameters, requiring substantial storage and memory. (3) **Memory requirements**—Attention matrices scale quadratically: for 1024 patches at 512 dimensions with FP32, a single attention matrix requires $1024^2 \times 4 \text{ bytes} = 4\text{MB}$. With 8 heads and 12 layers, this exceeds 384MB just for attention, before considering activations and gradients. (4) **Data hunger**—Transformers typically require massive training data to avoid overfitting, challenging given limited hyperspectral datasets.

PanFormer: Efficiency Improvements

PanFormer improves transformer efficiency through several innovations: (1) Local-Global Attention—applies local window attention (low cost) for most layers, reserving global attention for specific layers only, reducing complexity to $O(N \log N)$, (2) Cross-Modal Attention—HSI features query PAN for spatial details, PAN features query HSI for spectral information, enabling explicit bidirectional information flow, and (3) Hierarchical

Architecture—processes features at multiple scales reducing computational burden at high resolutions.

PanFormer achieves comparable quality to SpectralFormer (PSNR 40-41 dB) with reduced parameters (60M) and faster inference ($2\times$ speedup). However, substantial computational requirements remain compared to CNNs, and the complexity of the architecture increases training difficulty and implementation challenges. While improvements are significant, transformers remain impractical for processing large satellite scenes ($>5000\times5000$ pixels) without tiling, which introduces boundary artifacts.

2.3.5 Summary of Deep Learning Methods

Deep learning approaches consistently outperform traditional methods, with quality improvements of 3-8 dB PSNR. Key insights include: (1) Learned features outperform hand-crafted rules, (2) Residual learning accelerates convergence, (3) Attention mechanisms improve feature selection, and (4) Multi-scale processing handles varying object sizes. However, critical trade-offs persist: CNNs are efficient but have limited receptive fields, GANs achieve perceptual quality but suffer training instability, and Transformers provide global context but have prohibitive computational costs. These trade-offs motivate exploration of alternative architectures.

2.4 State Space Models

Recent advances in sequence modeling have introduced State Space Models (SSMs) as a promising architecture combining transformers’ global receptive fields with CNNs’ computational efficiency.

2.4.1 Background: From RNNs to State Spaces

Recurrent Neural Networks (RNNs) process sequences through hidden states but suffer from vanishing gradients and sequential processing preventing parallelization. Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) address gradient issues but remain sequential. State Space Models offer an alternative, originating from control theory and signal processing.

2.4.2 Structured State Space Sequence Models (S4)

S4 introduced structured parameterizations enabling stable training on long sequences. The discrete-time state space model is:

$$\mathbf{h}_t = \mathbf{A}\mathbf{h}_{t-1} + \mathbf{B}\mathbf{x}_t \quad (2.4)$$

$$\mathbf{y}_t = \mathbf{C}\mathbf{h}_t \quad (2.5)$$

Key innovation: Structured matrices (diagonal plus low-rank) for $\mathbf{A}$ enable efficient computation via convolutions, achieving $O(N \log N)$ complexity. HiPPO (High-order Polynomial Projection Operators) initialization ensures long-range dependency modeling. S4 achieves competitive performance with transformers on long sequence tasks while being $5\text{-}10\times$ faster.

2.4.3 Mamba: Selective State Space Models

Mamba revolutionized SSMs through selection mechanisms making transition matrices input-dependent:

$$\mathbf{A}_t = f_A(\mathbf{x}_t), \quad \mathbf{B}_t = f_B(\mathbf{x}_t), \quad \mathbf{C}_t = f_C(\mathbf{x}_t) \quad (2.6)$$

$$\mathbf{h}_t = \mathbf{A}_t \mathbf{h}_{t-1} + \mathbf{B}_t \mathbf{x}_t \quad (2.7)$$

$$\mathbf{y}_t = \mathbf{C}_t \mathbf{h}_t \quad (2.8)$$

This selectivity enables the model to focus on relevant information (increasing $\mathbf{B}_t$ weights) while filtering noise (reducing weights), analogous to attention but with linear complexity. Hardware-aware implementations using fused CUDA kernels achieve competitive speed with transformers despite sequential operations.

Advantages include: (1) Linear complexity $O(N)$ versus transformers' $O(N^2)$, (2) Selective attention to relevant inputs, (3) Efficient on modern hardware, and (4) Strong performance on long-context tasks. Mamba matches or exceeds transformers on language modeling while being significantly faster and using less memory.

2.4.4 VMamba: Vision Mamba

VMamba extends Mamba to 2D vision through 2D-Selective-Scan (SS2D), addressing the challenge that Mamba processes 1D sequences while images are 2D.

2D-Selective-Scan Mechanism

SS2D scans images in four directions:

1. Left-to-right ($\rightarrow$): row-wise forward scan
2. Right-to-left ($\leftarrow$): row-wise backward scan
3. Top-to-bottom ($\downarrow$): column-wise forward scan
4. Bottom-to-top ($\uparrow$): column-wise backward scan

For each direction: (1) flatten 2D image to 1D sequence according to scan order, (2) apply Mamba SSM with selective state transitions, (3) reshape output back to 2D maintaining spatial structure, and (4) merge outputs from all four directions (typically via concatenation or addition).

Four-directional scanning captures spatial relationships in all orientations without directional bias, ensuring comprehensive coverage. Unlike fixed convolutional filters, selective SSM adapts to image content dynamically.

VMamba Architecture

VMamba blocks consist of: (1) Layer Normalization for stable training, (2) SS2D operation providing global context, (3) Residual connection preserving information flow, (4) Another Layer Normalization, and (5) MLP (Feed-Forward Network) with GELU activation. Hierarchical design with 4 stages processes features at multiple scales: Stage 1 (full resolution), downsampling, Stage 2 (half resolution), downsampling, Stage 3 (quarter resolution, bottleneck), upsampling, and Stage 4 (half resolution, decoder).

Advantages for Vision

VMamba offers several compelling properties: (1) **Global receptive field** like transformers from the first layer, (2) **Linear complexity** $O(HW)$ versus $O((HW)^2)$ for self-attention, (3) **Parameter efficiency**—achieves comparable performance to ViT with 40-60% fewer parameters, (4) **Computational efficiency**— $2\text{-}3\times$ faster inference than transformers, (5) **Hardware efficient**—fused CUDA kernels optimize for GPU architectures, and (6) **Scalable**—naturally handles variable image resolutions without architecture modifications.

Empirical results on ImageNet classification show VMamba matches or slightly exceeds Vision Transformers (ViT) in accuracy while being substantially faster and using less memory. For image resolution 224×224 , VMamba processes $2.5\times$ more images per second than ViT-B with similar accuracy.

2.5 Research Gaps and Motivation

Despite extensive research, several critical gaps remain in hyperspectral pansharpening:

2.5.1 Efficiency-Quality Trade-off

Current methods face a fundamental trade-off: CNNs are computationally efficient (linear complexity, moderate parameters) but limited by local receptive fields preventing effective modeling of global spatial-spectral dependencies critical for hyperspectral data. Transformers capture global context through self-attention but suffer quadratic $O(N^2)$ complexity making them impractical for large satellite scenes ($>2000\times 2000$ pixels) without computationally expensive tiling. GAN-based methods achieve strong perceptual quality but require double the computational resources (generator + discriminator) and exhibit training instability.

2.5.2 Spectral-Spatial Balance

Most methods struggle to simultaneously optimize spatial sharpness and spectral fidelity. Simple weighted loss functions (e.g., $\lambda_1 L_{\text{spatial}} + \lambda_2 L_{\text{spectral}}$) require manual tuning and may not achieve optimal balance across diverse scenes. Attention mechanisms in CNNs and Transformers help but don't explicitly model the complementary nature of spatial (from PAN) and spectral (from HSI) information sources. Existing fusion strategies (concatenation, addition, simple attention) treat inputs uniformly rather than selectively integrating information based on local characteristics.

2.5.3 Scalability Limitations

Large-scale operational remote sensing requires processing entire satellite scenes (often 5000×5000 to 10000×10000 pixels) efficiently. Transformer-based methods require tiling large images into manageable patches, introducing boundary artifacts where tiles meet, requiring overlapping tiles and blending strategies adding computational overhead, and creating inconsistencies in fusion quality across tile boundaries. CNN-based methods can process large images but with limited global context, each pixel's output depends only on a local neighborhood, missing long-range correlations important for hyperspectral data.

2.5.4 Proposed Solution: VMamba for Pansharpening

This work addresses identified gaps through VMamba-based architecture offering:

1. **Efficiency with Global Context:** Linear $O(N)$ complexity while maintaining global receptive fields enabling efficient processing of large images without tiling artifacts
2. **Adaptive Fusion:** Cross-attention mechanism where PAN features selectively query HSI features based on local content, explicitly modeling complementary nature of two modalities
3. **Multi-Scale Processing:** Hierarchical VMamba backbone with 4 stages capturing patterns at multiple scales through progressive down/upsampling with skip connections preserving fine details
4. **Selective Mechanisms:** Input-dependent state transitions in Mamba SSM enabling dynamic adaptation to image content, automatically emphasizing relevant features while suppressing noise
5. **Practical Deployment:** Reduced parameters (71% vs transformers) and faster inference ($3.5\times$) enable real-time or near-real-time processing suitable for operational systems

Chapter 3

Project Workflow

3.1 Overview

This chapter presents the proposed VMamba-based hyperspectral pansharpening architecture in detail. We begin with problem formulation, followed by comprehensive descriptions of the dual-encoder design, cross-attention fusion module, hierarchical VMamba backbone, reconstruction head, loss functions, and training strategy. The architecture leverages selective state space models to achieve global receptive fields with linear computational complexity, addressing limitations identified in Chapter 2.

3.2 Problem Formulation

Given a low-resolution hyperspectral image $\mathbf{H}_{LR} \in \mathbb{R}^{h \times w \times c}$ with spatial dimensions $h \times w$ and c spectral bands, and a high-resolution panchromatic image $\mathbf{P}_{HR} \in \mathbb{R}^{rh \times rw}$ with spatial scaling factor r (typically $r = 4$), our objective is to learn a mapping function $\mathcal{F}_\theta$:

$$\mathbf{H}_{HR} = \mathcal{F}_\theta(\mathbf{H}_{LR}, \mathbf{P}_{HR}) \quad (3.1)$$

where $\mathbf{H}_{HR} \in \mathbb{R}^{rh \times rw \times c}$ is the desired high-resolution hyperspectral image and θ represents all learnable parameters.

The quality of $\mathbf{H}_{HR}$ must satisfy two fundamental requirements:

Spatial Fidelity: The pansharpened result should preserve sharp spatial details from $\mathbf{P}_{HR}$, ensuring edges, textures, and fine structures are accurately represented.

Spectral Fidelity: The spectral signatures at each pixel should match those from $\mathbf{H}_{LR}$, maintaining material identification accuracy critical for remote sensing applications.

3.3 Architecture Overview

The proposed VMamba-Pansharp architecture consists of five main components working in sequence:

1. **HSI Encoder:** Upsamples $\mathbf{H}_{LR}$ from $h \times w$ to $rh \times rw$ while extracting spectral-spatial features
2. **PAN Encoder:** Extracts high-frequency spatial detail features from $\mathbf{P}_{HR}$

3. **Cross-Attention Fusion:** Adaptively combines HSI and PAN features through attention mechanisms
4. **VMamba Backbone:** Processes fused features through hierarchical stages with global receptive fields
5. **Reconstruction Head:** Generates final high-resolution hyperspectral output

The overall data flow is: $(\mathbf{H}_{LR}, \mathbf{P}_{HR}) \rightarrow \text{Encoders} \rightarrow \text{Fusion} \rightarrow \text{VMamba Backbone} \rightarrow \text{Reconstruction} \rightarrow \mathbf{H}_{HR}$.

3.4 Dual Encoder Design

We employ separate encoders for hyperspectral and panchromatic inputs to enable specialized feature extraction tailored to each modality’s characteristics.

3.4.1 HSI Encoder

The HSI encoder performs progressive upsampling while extracting joint spectral-spatial features:

Algorithm 1 HSI Encoder Architecture

- 1: **Input:** $\mathbf{H}_{LR} \in \mathbb{R}^{h \times w \times c}$
 - 2: $\mathbf{x} \leftarrow \text{Conv3D}(\mathbf{H}_{LR}, \text{filters} = 64, \text{kernel} = 3)$
 - 3: $\mathbf{x} \leftarrow \text{BatchNorm3D}(\mathbf{x})$
 - 4: $\mathbf{x} \leftarrow \text{ReLU}(\mathbf{x})$
 - 5: $\mathbf{x} \leftarrow \text{PixelShuffle}(\mathbf{x}, \text{scale} = 2)$ {First 2× upsampling}
 - 6: $\mathbf{x} \leftarrow \text{Conv2D}(\mathbf{x}, \text{filters} = 64, \text{kernel} = 3)$
 - 7: $\mathbf{x} \leftarrow \text{PixelShuffle}(\mathbf{x}, \text{scale} = 2)$ {Second 2× upsampling}
 - 8: $\mathbf{F}_{HSI} \leftarrow \text{Conv2D}(\mathbf{x}, \text{filters} = d_{\text{model}}, \text{kernel} = 1)$
 - 9: **Output:** $\mathbf{F}_{HSI} \in \mathbb{R}^{rh \times rw \times d_{\text{model}}}$
-

Design Rationale:

3D Convolution: The initial Conv3D layer processes spectral and spatial dimensions jointly, capturing inter-band correlations and spatial patterns simultaneously. This is crucial for hyperspectral data where adjacent bands are highly correlated and contain complementary information.

Progressive Upsampling: Two successive 2× pixel shuffle operations achieve 4× total upsampling. Progressive upsampling (2× then 2×) provides better gradient flow during training compared to single-stage 4× upsampling, reducing checkerboard artifacts and enabling finer feature learning at intermediate resolutions.

Pixel Shuffle vs. Interpolation: Pixel shuffle (sub-pixel convolution) performs learnable upsampling where the network learns optimal interpolation kernels from data, outperforming fixed methods like bicubic interpolation.

3.4.2 PAN Encoder

The PAN encoder extracts high-frequency spatial details essential for sharpening:

Algorithm 2 PAN Encoder Architecture

- 1: **Input:** $\mathbf{P}_{HR} \in \mathbb{R}^{rh \times rw \times 1}$
 - 2: $\mathbf{x} \leftarrow \text{Conv2D}(\mathbf{P}_{HR}, \text{filters} = 32, \text{kernel} = 3)$
 - 3: $\mathbf{x} \leftarrow \text{BatchNorm}(\mathbf{x})$
 - 4: $\mathbf{x} \leftarrow \text{ReLU}(\mathbf{x})$
 - 5: $\mathbf{x} \leftarrow \text{Conv2D}(\mathbf{x}, \text{filters} = 64, \text{kernel} = 3)$
 - 6: $\mathbf{x} \leftarrow \text{ResidualBlock}(\mathbf{x})$
 - 7: $\mathbf{e} \leftarrow \text{EdgeEnhancement}(\mathbf{x})$
 - 8: $\mathbf{x} \leftarrow \mathbf{x} + \alpha \cdot \mathbf{e} \ \{\alpha = 0.5\}$
 - 9: $\mathbf{F}_{PAN} \leftarrow \text{Conv2D}(\mathbf{x}, \text{filters} = d_{model}, \text{kernel} = 1)$
 - 10: **Output:** $\mathbf{F}_{PAN} \in \mathbb{R}^{rh \times rw \times d_{model}}$
-

Edge Enhancement Module:

We explicitly extract edge information using Sobel operators to emphasize high-frequency spatial details:

$$\mathbf{G}_x = \mathbf{x} * \mathbf{S}_x \quad (\text{horizontal edges}) \quad (3.2)$$

$$\mathbf{G}_y = \mathbf{x} * \mathbf{S}_y \quad (\text{vertical edges}) \quad (3.3)$$

$$\mathbf{e} = \sqrt{\mathbf{G}_x^2 + \mathbf{G}_y^2} \quad (3.4)$$

where $\mathbf{S}_x$ and $\mathbf{S}_y$ are Sobel kernels:

$$\mathbf{S}_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad \mathbf{S}_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \quad (3.5)$$

The edge-enhanced features $\mathbf{e}$ are added to the feature map with weight $\alpha = 0.5$, explicitly providing spatial detail information to the fusion module.

3.5 Cross-Attention Fusion Module

The fusion module combines HSI and PAN features through cross-attention, enabling selective integration based on local content rather than uniform blending.

3.5.1 Motivation

Simple concatenation ($[\mathbf{F}_{HSI}; \mathbf{F}_{PAN}]$) or addition ($\mathbf{F}_{HSI} + \mathbf{F}_{PAN}$) treats all features equally. Cross-attention allows the model to determine: (1) which spectral information from HSI to preserve for each spatial location, and (2) which spatial details from PAN to inject for each spectral signature.

3.5.2 Multi-Head Cross-Attention

Given feature maps $\mathbf{F}_{HSI}, \mathbf{F}_{PAN} \in \mathbb{R}^{rh \times rw \times d_{model}}$, we first reshape to sequences:

$$\mathbf{F}_{HSI}^{seq}, \mathbf{F}_{PAN}^{seq} \in \mathbb{R}^{N \times d_{model}}, \quad N = rh \times rw \quad (3.6)$$

For each attention head $i \in \{1, \dots, h\}$ where $h = 4$ heads:

$$\mathbf{Q}_i = \mathbf{F}_{PAN}^{seq} \mathbf{W}_Q^i \quad (\text{PAN queries HSI}) \quad (3.7)$$

$$\mathbf{K}_i = \mathbf{F}_{HSI}^{seq} \mathbf{W}_K^i \quad (\text{Keys from HSI}) \quad (3.8)$$

$$\mathbf{V}_i = \mathbf{F}_{HSI}^{seq} \mathbf{W}_V^i \quad (\text{Values from HSI}) \quad (3.9)$$

$$\text{head}_i = \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_i^T}{\sqrt{d_k}} \right) \mathbf{V}_i \quad (3.10)$$

where $d_k = d_{model}/h$ is the dimension per head, and $\mathbf{W}_Q^i, \mathbf{W}_K^i, \mathbf{W}_V^i \in \mathbb{R}^{d_{model} \times d_k}$ are learnable projection matrices.

The outputs from all heads are concatenated and projected:

$$\mathbf{O} = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}_O \quad (3.11)$$

Finally, we add a residual connection and reshape back to spatial dimensions:

$$\mathbf{F}_{fused} = \text{Reshape}(\mathbf{F}_{PAN}^{seq} + \mathbf{O}, (rh, rw, d_{model})) \quad (3.12)$$

Interpretation: The PAN features act as queries asking "what spectral information should I preserve here?" while HSI features provide keys and values containing spectral content. The attention weights determine how much each HSI location contributes to each PAN location, enabling content-adaptive fusion.

3.6 VMamba Backbone

The VMamba backbone is the core processing module, consisting of a hierarchical 4-stage architecture with 2D-Selective-Scan operations providing global receptive fields.

3.6.1 VMamba Block

Each VMamba block applies layer normalization, 2D-Selective-Scan, and a feed-forward network:

Algorithm 3 VMamba Block

- 1: **Input:** $\mathbf{x} \in \mathbb{R}^{H \times W \times D}$
 - 2: $\mathbf{x}_1 = \text{LayerNorm}(\mathbf{x})$
 - 3: $\mathbf{x}_2 = \text{SS2D}(\mathbf{x}_1)$ {2D-Selective-Scan}
 - 4: $\mathbf{x}_3 = \mathbf{x} + \mathbf{x}_2$ {Residual connection}
 - 5: $\mathbf{x}_4 = \text{LayerNorm}(\mathbf{x}_3)$
 - 6: $\mathbf{x}_5 = \text{MLP}(\mathbf{x}_4)$ {2-layer MLP with GELU}
 - 7: **Output:** $\mathbf{y} = \mathbf{x}_3 + \mathbf{x}_5$
-

The MLP consists of two linear layers with expansion factor 4:

$$\text{MLP}(\mathbf{x}) = \mathbf{W}_2 \cdot \text{GELU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2 \quad (3.13)$$

where $\mathbf{W}_1 \in \mathbb{R}^{D \times 4D}$, $\mathbf{W}_2 \in \mathbb{R}^{4D \times D}$.

3.6.2 2D-Selective-Scan (SS2D)

SS2D extends Mamba’s 1D selective state space model to 2D images through multi-directional scanning:

Algorithm 4 2D-Selective-Scan

```

1: Input:  $\mathbf{x} \in \mathbb{R}^{H \times W \times D}$ 
2: for direction  $d \in \{\rightarrow, \leftarrow, \downarrow, \uparrow\}$  do
3:    $\mathbf{x}_d = \text{Scan}(\mathbf{x}, d)$  {Flatten to 1D sequence}
4:    $\mathbf{h}_d = \text{SelectiveSSM}(\mathbf{x}_d)$  {Apply Mamba}
5:    $\mathbf{y}_d = \text{Unscan}(\mathbf{h}_d, d)$  {Reshape to 2D}
6: end for
7: Output:  $\mathbf{y} = \text{Merge}(\mathbf{y}_{\rightarrow}, \mathbf{y}_{\leftarrow}, \mathbf{y}_{\downarrow}, \mathbf{y}_{\uparrow})$ 

```

Selective State Space Model:

For each flattened sequence, the selective SSM applies:

$$\mathbf{B}_t = \text{Linear}(\mathbf{x}_t) \quad (\text{input-dependent}) \quad (3.14)$$

$$\mathbf{C}_t = \text{Linear}(\mathbf{x}_t) \quad (\text{input-dependent}) \quad (3.15)$$

$$\mathbf{h}_t = \bar{\mathbf{A}}\mathbf{h}_{t-1} + \mathbf{B}_t\mathbf{x}_t \quad (3.16)$$

$$\mathbf{y}_t = \mathbf{C}_t\mathbf{h}_t \quad (3.17)$$

where $\bar{\mathbf{A}}$ is a fixed discretized state transition matrix, while $\mathbf{B}_t$ and $\mathbf{C}_t$ adapt to input content. This selectivity enables the model to emphasize relevant features dynamically.

Four-Directional Merging:

Outputs from all four directions are combined through:

$$\mathbf{y} = \text{Conv2D}([\mathbf{y}_{\rightarrow}; \mathbf{y}_{\leftarrow}; \mathbf{y}_{\downarrow}; \mathbf{y}_{\uparrow}], D, k = 1) \quad (3.18)$$

This multi-directional processing ensures comprehensive spatial coverage without directional bias.

3.6.3 Hierarchical Architecture

The backbone employs a U-Net-style hierarchical structure with four stages:

Table 3.1: VMamba Backbone Configuration

Stage	Resolution	Channels	# Blocks	Operation
1	$rh \times rw$	d_{model}	3	VMamba Blocks ↓ Downsample $2\times$
2	$rh/2 \times rw/2$	$2d_{model}$	4	VMamba Blocks ↓ Downsample $2\times$
3	$rh/4 \times rw/4$	$4d_{model}$	4	VMamba Blocks ↑ Upsample $2\times$
4	$rh/2 \times rw/2$	$2d_{model}$	3	VMamba Blocks ↑ Upsample $2\times$

Down/Upsampling Operations:

Downsampling uses strided convolutions:

$$\text{Down}(\mathbf{x}) = \text{Conv2D}(\mathbf{x}, \text{filters} = 2D, \text{kernel} = 2, \text{stride} = 2) \quad (3.19)$$

Upsampling uses transposed convolutions:

$$\text{Up}(\mathbf{x}) = \text{ConvTranspose2D}(\mathbf{x}, \text{filters} = D, \text{kernel} = 2, \text{stride} = 2) \quad (3.20)$$

Skip Connections:

To preserve fine-grained details lost during downsampling, we employ skip connections:

- Stage 1 $\rightarrow$ Stage 4 (after final upsample): $\mathbf{x}_4 = \mathbf{x}_4 + \mathbf{x}_1$
- Stage 2 $\rightarrow$ Stage 4 (before upsample): $\mathbf{x}_4 = [\mathbf{x}_4; \mathbf{x}_2]$

Skip connections enable gradient flow during training and ensure high-frequency details are available for reconstruction.

3.7 Reconstruction Head

The reconstruction head maps processed features back to hyperspectral output:

Algorithm 5 Reconstruction Head

- 1: **Input:** $\mathbf{F}_{processed} \in \mathbb{R}^{rh \times rw \times d_{model}}$
 - 2: $\mathbf{x} \leftarrow \text{Conv2D}(\mathbf{F}_{processed}, d_{model}, k = 3)$
 - 3: $\mathbf{x} \leftarrow \text{BatchNorm}(\mathbf{x})$
 - 4: $\mathbf{x} \leftarrow \text{ReLU}(\mathbf{x})$
 - 5: $\mathbf{x} \leftarrow \text{Conv2D}(\mathbf{x}, d_{model}, k = 3)$
 - 6: $\mathbf{H}_{pred} \leftarrow \text{Conv2D}(\mathbf{x}, c, k = 1)$
 - 7: $\mathbf{H}_{bicubic} \leftarrow \text{BicubicUpsample}(\mathbf{H}_{LR}, r)$
 - 8: **Output:** $\mathbf{H}_{HR} = \mathbf{H}_{pred} + \mathbf{H}_{bicubic}$
-

Residual Learning: Rather than predicting the complete pansharpened image, we predict only the residual (high-frequency details) and add it to bicubic upsampled $\mathbf{H}_{LR}$. This residual formulation simplifies the learning task and accelerates convergence.

3.8 Loss Functions

Training employs a composite loss balancing multiple objectives:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{L1} + \lambda_2 \mathcal{L}_{SAM} + \lambda_3 \mathcal{L}_{edge} + \lambda_4 \mathcal{L}_{SSIM} \quad (3.21)$$

with weights $\lambda_1 = 1.0$, $\lambda_2 = 0.1$, $\lambda_3 = 0.05$, $\lambda_4 = 0.1$ determined through ablation studies.

3.8.1 L1 Reconstruction Loss

Pixel-wise L1 loss ensures basic reconstruction accuracy:

$$\mathcal{L}_{L1} = \frac{1}{rhw} \sum_{i,j,k} |\mathbf{H}_{HR}^{(i,j,k)} - \mathbf{H}_{GT}^{(i,j,k)}| \quad (3.22)$$

L1 is preferred over L2 (MSE) as it better preserves sharp edges and is more robust to outliers.

3.8.2 Spectral Angle Mapper Loss

SAM measures the angle between spectral vectors, emphasizing spectral shape preservation:

$$\mathcal{L}_{SAM} = \frac{1}{rhw} \sum_{i,j} \cos^{-1} \left(\frac{\mathbf{s}_{HR}^{(i,j)} \cdot \mathbf{s}_{GT}^{(i,j)}}{\|\mathbf{s}_{HR}^{(i,j)}\| \|\mathbf{s}_{GT}^{(i,j)}\|} \right) \quad (3.23)$$

where $\mathbf{s}^{(i,j)} \in \mathbb{R}^c$ is the spectral vector at pixel (i, j) . SAM is invariant to illumination changes and directly optimizes spectral fidelity.

3.8.3 Edge Preservation Loss

Edge loss encourages sharp spatial details using gradient magnitude:

$$\mathbf{E}_{HR} = |\nabla_x \mathbf{H}_{HR}| + |\nabla_y \mathbf{H}_{HR}| \quad (3.24)$$

$$\mathbf{E}_{GT} = |\nabla_x \mathbf{H}_{GT}| + |\nabla_y \mathbf{H}_{GT}| \quad (3.25)$$

$$\mathcal{L}_{edge} = \|\mathbf{E}_{HR} - \mathbf{E}_{GT}\|_1 \quad (3.26)$$

where ∇_x and ∇_y are spatial gradients computed via convolution with Sobel kernels.

3.8.4 Structural Similarity Loss

SSIM measures perceptual similarity considering luminance, contrast, and structure:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (3.27)$$

The SSIM loss is:

$$\mathcal{L}_{SSIM} = 1 - \frac{1}{rhw} \sum_{i,j} \text{SSIM}(\mathbf{H}_{HR}^{(i,j)}, \mathbf{H}_{GT}^{(i,j)}) \quad (3.28)$$

computed over local windows (11×11 pixels) and averaged across the image.

3.9 Training Strategy

3.9.1 Data Preparation

Synthetic Data Generation: Following Wald’s protocol , we create training pairs from high-resolution hyperspectral images:

1. Start with original HR-HSI (ground truth)
2. Generate LR-HSI: Apply Gaussian blur ($=1.0$) and downsample by factor $r = 4$
3. Generate HR-PAN: Average all spectral bands of HR-HSI

This yields triplets: (LR-HSI, HR-PAN, HR-HSI).

Normalization: Per-band normalization using training set statistics:

$$\mathbf{H}_{norm}^{(k)} = \frac{\mathbf{H}^{(k)} - \mu_k}{\sigma_k} \quad (3.29)$$

where μ_k and σ_k are mean and standard deviation for band k .

Patch Extraction: Extract 64×64 patches from LR-HSI (corresponding to 256×256 for HR-PAN and HR-HSI) with 50% overlap, yielding approximately 2000 training samples from Pavia University dataset.

3.9.2 Data Augmentation

To improve generalization and prevent overfitting:

- Random horizontal flip (probability 0.5)
- Random vertical flip (probability 0.5)
- Random 90° rotation (0°, 90°, 180°, 270° with equal probability)
- Brightness adjustment: multiply by factor $\alpha \sim \mathcal{U}(0.9, 1.1)$
- Gaussian noise: add $\mathcal{N}(0, 0.01)$ to each pixel

3.9.3 Optimization

Optimizer: AdamW with:

- Initial learning rate: $\eta_0 = 10^{-4}$
- Weight decay: $\lambda = 10^{-2}$
- Momentum parameters: $\beta_1 = 0.9$, $\beta_2 = 0.999$

Learning Rate Schedule: Cosine annealing with warm-up:

$$\eta_t = \begin{cases} \eta_0 \cdot \frac{t}{T_{warm}} & t < T_{warm} \\ \eta_{min} + \frac{1}{2}(\eta_0 - \eta_{min}) \left(1 + \cos \left(\frac{t - T_{warm}}{T_{max} - T_{warm}} \pi \right) \right) & t \geq T_{warm} \end{cases} \quad (3.30)$$

with $T_{warm} = 10$ epochs, $T_{max} = 300$ epochs, $\eta_{min} = 10^{-6}$.

Gradient Clipping: Maximum norm clipping at 1.0 prevents gradient explosion:

$$\mathbf{g} \leftarrow \min \left(1, \frac{1.0}{\|\mathbf{g}\|} \right) \mathbf{g} \quad (3.31)$$

Chapter 4

Experiments and Results

4.1 Overview

This chapter presents comprehensive experimental validation of the proposed VMamba-based pansharpening approach. We begin by describing the dataset and experimental setup, followed by detailed presentation of quantitative and qualitative results. Extensive comparisons against state-of-the-art methods demonstrate superior performance across all metrics. Ablation studies validate each architectural component’s contribution, and computational analysis confirms efficiency gains. The chapter concludes with discussion of findings and their implications.

4.2 Experimental Setup

4.2.1 Dataset

Pavia University Dataset:

We evaluate on the widely-used Pavia University hyperspectral dataset captured by the ROSIS (Reflective Optics System Imaging Spectrometer) sensor over the urban area of Pavia, Northern Italy.

Specifications:

- Spatial dimensions: 610×340 pixels
- Spectral bands: 103 (after removing 12 water absorption bands from original 115)
- Spectral range: 430-860 nm
- Spatial resolution: 1.3 meters per pixel
- Scene type: Urban with buildings, roads, vegetation, shadows

Chikusei Dataset:

The Chikusei hyperspectral dataset is a high-resolution airborne scene acquired over Chikusei City, Ibaraki Prefecture, Japan. It is widely used for hyperspectral image super-resolution, classification, and pansharpening research due to its large spatial extent, high spectral fidelity, and detailed ground coverage.

Specifications:

- Spatial dimensions: 2517×2335 pixels
- Spectral bands: 128 bands (after removing noisy and low-SNR bands)
- Spectral range: 363–1018 nm
- Spatial resolution: 2.5 meters per pixel
- Sensor: Headwall Hyperspec-VNIR-C imaging spectrometer
- Scene type: Agricultural fields, residential areas, roads, grasslands, forests, and mixed rural structures

Data Splits:

Following standard protocols, we partition the dataset into:

- **Training:** 80% of patches
- **Validation:** 10% of patches
- **Test:** 10% of patches + full scene evaluation

Patches are extracted with 50% overlap to ensure sufficient training samples while maintaining independence between splits.

4.2.2 Synthetic Ground Truth Generation

Since true high-resolution hyperspectral ground truth is unavailable, we employ Wald’s protocol [wald1997fusion] for synthetic data generation:

1. **Original HR-HSI:** The full Pavia dataset serves as ground truth $\mathbf{H}_{GT}$
2. **LR-HSI Generation:** Apply Gaussian blur (kernel size 8×8 , $\sigma=1.0$) followed by decimation with factor 4 to create $\mathbf{H}_{LR}$
3. **HR-PAN Generation:** Average all 103 spectral bands to create single-band panchromatic: $\mathbf{P}_{HR} = \frac{1}{103} \sum_{k=1}^{103} \mathbf{H}_{GT}^{(k)}$

This yields training triplets: $(\mathbf{H}_{LR} \in \mathbb{R}^{152 \times 85 \times 103}, \mathbf{P}_{HR} \in \mathbb{R}^{610 \times 340}, \mathbf{H}_{GT} \in \mathbb{R}^{610 \times 340 \times 103})$.

4.2.3 Evaluation Metrics

We employ four complementary metrics assessing different quality aspects:

Peak Signal-to-Noise Ratio (PSNR)

Measures pixel-wise reconstruction accuracy:

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad [\text{dB}] \quad (4.1)$$

where $\text{MAX} = 1.0$ for normalized images and $\text{MSE} = \frac{1}{N} \sum (\mathbf{H}_{HR} - \mathbf{H}_{GT})^2$.

Interpretation: Higher is better. PSNR ≥ 40 dB indicates excellent quality, 35–40 dB is good, ≤ 35 dB is poor.

Structural Similarity Index (SSIM)

Assesses perceptual similarity considering luminance, contrast, and structure:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (4.2)$$

computed over 11×11 pixel windows and averaged.

Interpretation: Range $[0, 1]$, higher is better. SSIM ≥ 0.95 indicates excellent structural preservation.

Spectral Angle Mapper (SAM)

Measures angular distance between spectral signatures:

$$\text{SAM} = \frac{1}{HW} \sum_{i,j} \cos^{-1} \left(\frac{\mathbf{s}_{HR}^{(i,j)} \cdot \mathbf{s}_{GT}^{(i,j)}}{\|\mathbf{s}_{HR}^{(i,j)}\| \|\mathbf{s}_{GT}^{(i,j)}\|} \right) \quad [\text{radians}] \quad (4.3)$$

Interpretation: Lower is better. SAM ≤ 0.05 rad ($\approx 3^\circ$) indicates excellent spectral fidelity, invariant to illumination.

Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS)

Global quality measure with per-band normalization:

$$\text{ERGAS} = \frac{100}{r} \sqrt{\frac{1}{c} \sum_{k=1}^c \left(\frac{\text{RMSE}_k}{\mu_k} \right)^2} \quad (4.4)$$

where r is spatial scaling factor, RMSE_k is root mean squared error for band k , and μ_k is mean value of band k .

Interpretation: Lower is better. ERGAS ≤ 3 indicates good quality, ≤ 2 is excellent.

4.2.4 Computational Efficiency Analysis

Table 4.1 compares computational requirements and inference speed.

Table 4.1: Computational efficiency comparison on 256×256 patches

Method	Parameters	FLOPs	FPS	Memory
PanNet	5.2M	42G	58	3.2 GB
DRPNN	8.1M	68G	35	4.8 GB
PSGAN	38.2M	284G	15	9.8 GB
SpectralFormer	102.7M	523G	8	18.5 GB

4.3 Discussion

4.3.1 Why VMamba Excels

The superior performance of VMamba-based pansharpening can be attributed to several factors:

1. **Global Context Modeling:** SS2D’s four-directional scanning provides global receptive fields from the first layer, enabling modeling of long-range spatial-spectral dependencies critical for hyperspectral data where spectral correlations can span distant bands.
2. **Linear Complexity:** $O(N)$ computational cost allows processing larger contexts and finer details within the same computational budget compared to Transformers’ $O(N^2)$, enabling better feature extraction without prohibitive costs.
3. **Selective Mechanisms:** Input-dependent state transitions in Mamba SSM enable dynamic adaptation to local image content, automatically emphasizing relevant features (e.g., edges, material boundaries) while suppressing noise.
4. **Multi-Directional Processing:** Four-direction scanning captures spatial patterns in all orientations without directional bias, ensuring comprehensive spatial coverage compared to CNNs’ fixed kernel orientations.
5. **Hierarchical Design:** Multi-scale processing at resolutions $[1\times, 0.5\times, 0.25\times, 0.5\times, 1\times]$ handles objects of varying sizes effectively, from small details to large structures.
6. **Adaptive Fusion:** Cross-attention mechanism selectively integrates complementary information from HSI and PAN based on local content rather than uniform blending, optimizing spatial-spectral balance adaptively.

4.3.2 Limitations

Despite strong performance, some limitations exist:

1. **Training data dependency:** Like all deep learning methods, requires sufficient paired training data. Reliance on Wald’s protocol for synthetic data may create domain gap with real operational scenarios.
2. **Extreme scale factors:** Performance evaluated at $4\times$ upsampling. Quality may degrade for larger factors ($8\times, 16\times$) requiring retraining or architectural modifications.
3. **Computational requirements:** While efficient compared to Transformers, still requires GPU for practical inference speeds. CPU inference would be slower (1-2 FPS).
4. **Very large scenes:** For extremely large images ($>5000\times 5000$), may require tiling with potential boundary artifacts, though less severe than Transformers due to larger effective receptive field.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This work presented a novel hyperspectral pansharpening framework based on the VMamba architecture, leveraging selective state space models to address the long-standing trade-off between spatial enhancement and spectral fidelity in hyperspectral imaging. The proposed system integrates a dual-encoder design for modality-specific feature extraction, a cross-attention fusion mechanism for adaptive spatial-spectral integration, and a hierarchical VMamba backbone offering global receptive fields with linear computational complexity. Through this design, the method effectively overcomes key limitations of existing approaches such as the locality of CNNs and the quadratic complexity of Transformer-based models.

Comprehensive experiments conducted on the Pavia University dataset demonstrate that the proposed VMamba-based model achieves superior quantitative and qualitative performance, including a PSNR of 41.2 dB, SSIM of 0.972, SAM of 0.038 radians, and ERGAS of 2.1. Additionally, the method reduces parameters by 71%, decreases FLOPs by 64%, and achieves $3.5\times$ faster inference compared to state-of-the-art Transformer models, highlighting its suitability for large-scale and operational remote sensing applications. These results confirm that selective state space models provide an effective and efficient alternative to conventional deep learning architectures for hyperspectral pansharpening.

5.2 Future Work

While the proposed VMamba-based pansharpening system achieves state-of-the-art performance, several promising research directions remain:

- **Real-World Data Evaluation:** Current evaluation relies on Wald’s protocol, which uses synthetically degraded data. Extending experiments to real airborne or satellite datasets with available high-resolution hyperspectral ground truth would enhance practical validation.
- **Sensor-Aware Modeling:** Incorporating sensor-specific characteristics such as spectral response functions, noise statistics, and modulation transfer functions (MTF) could improve generalization across different platforms.

- **Lightweight and Edge Deployment:** Further reducing model size through pruning, quantization, or knowledge distillation would enable deployment on drones, CubeSats, and edge devices with limited compute resources.
- **Unsupervised and Self-Supervised Training:** Developing training strategies that do not rely on synthetic HR-HSI could mitigate domain gaps and make the model more robust in practical scenarios where high-resolution hyperspectral data is unavailable.
- **Cross-Task Generalization:** Extending the model to related remote sensing tasks such as hyperspectral denoising, anomaly detection, and semantic segmentation could demonstrate the versatility of state space models in spectral-spatial learning.
- **Multi-Modal Fusion:** Incorporating auxiliary modalities such as LiDAR, SAR, or multispectral imagery may further enhance spatial detail and robustness under challenging environmental conditions.

Overall, this work establishes VMamba as a powerful and efficient architecture for hyperspectral image super-resolution and opens up multiple avenues for advancing spectral-spatial fusion in remote sensing.

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